





ARBA MINCH UNIVERSITY
COLLEGE OF NATURAL SCIENCES
DEPARTMENT OF STATISTICS

SAMPLE MODEL EXAM FOR GRADUATING CLASS STATISTICS

DEPARTMENT STUDENTS

EXAM DATE:- 31/03/2023

MAX. MARKS:-100%

STUDENT NAME:

TIME ALLOWED:- 3:00 HRS

ID.NO. NSR1796/12

General Instructions

1. Before writing exam, **switch off** your mobile and make sure this cover page has department stamp on it.
2. Make sure that the examination has 120 multiple choice questions on 20 pages.
3. Attempt all questions and write your answers clearly on the separate answer sheet
4. Use Calculators and other allowed stationary materials of your own. Don't share with others.
5. Any attempt to cheat will result in serious penalty and unreadable answers will not be marked.

CHOOSE THE BEST ANSWER

(Each 1 pt=120%)

1. Suppose that you are asked ten of your classmates about their height. On the basis of this information, you computed the average and stated that the average height of all students in your college is 70 inches. This is an example of _____
 A. Descriptive statistics
 B. Inferential Statistics
 C. Parameter
 D. Population
 E. None of the above
2. Questionnaire method is used to collect _____
 A. Primary data
 B. Secondary Data
 C. A & B
 D. None of the above
3. The following frequency distribution represents the result of 60 Statistics 3rd year students in the year 2009 E.C at AMU on Basic Statistics course out of 100

Mark Scored	30-39	40-49	50-59	60-69	70-79	80-89
Frequency	5	10	10	15	15	5

Based on frequency distribution, how many students Scored at least 70?

- A. 55
 B. 20
 C. 15
 D. 5
 E. None of the above
 4. Which of the following measure of central tendency is highly affected by extreme observations?
 A. Mean
 B. Median
 C. Mode
 D. All of the above
 5. Which measure of central tendency could be computed on a variable that is measured on a nominal scale of measurement?
 A. Mean
 B. Median
 C. Mode
 D. None of the above
 6. An exam is given and the instructor noted that the mean time to finish the exam was 37.5 minutes with a standard deviation of 10.5. The instructor decided to add one more question to the exam. The exam now takes each of the students another 5 minutes. What will be the value of the new mean and new standard deviation respectively?
 A. 37.5, 10.5
 B. 42.5, 10.5
 C. 42.5, 110.25
 D. 187.5, 52.5
 E. None of the above
- $37.5 + 5 = 42.5$
 $10.5 \times 2 = 21$
 $21 + 92.5 = 113.5$
7. If the value of first quartile for the ordered wage of 20 employees is found to be \$2500 per month. What can we say about this value?
 A. 50% of employees are paid less than or equal to \$2500.
 B. 25% of employees are paid greater than or equal to \$2500.
 C. 25% of employees are paid less than or equal to \$2500.
 D. 20% of employees are paid less than or equal to \$2500.
 8. Which of the following is true statement?
 A. Variance is relative measure of dispersion.
 B. Mean deviation is the arithmetic mean of square deviation about the mean

Graduating class Statistics department Students

- C. When the values are uniformly distributed around the mean a distribution is said to be symmetrical
 D. All of the above
9. _____ is measure of variation used to measure the distance of individual observation from their average in the unit of standard deviation and indicates relative performance
 A. Coefficient-variation
 B. Standard score
 C. Variance
 D. Quartile deviation
10. If the range and relative range of a series are 4 and 0.25 respectively, then the value of largest and smallest observations, respectively are _____
 A. 20 and 16
 B. 16 and 12
 C. 10 and 6
 D. None of the above

11. One hundred randomly selected Arba Minch university students in 2009 E.C were asked whether or not they have ever visited the Crocodiles Ranch. The following table gives a two-way classification of the responses.

	Have visited(V)	Have never visited(NV)	Total
Male(M)	15	40	55
Female(F)	10	35	45
Total	25	75	100

If one student is selected at random from 100 students, the probability that this student has never visited Crocodiles Ranch is _____

- A. 0.15
 B. 0.25
 C. 0.55
 D. 0.75

12. Refer question 11, the probability that the selected student is a male given that the student has never visited Crocodiles Ranch is _____

- A. 0.25
 B. 0.375
 C. 0.533
 D. None of the above

13. A researcher divided subjects into two groups according to gender and then selected members from each group for his/her sample. The sampling method used was _____
 A. Simple Random sampling
 B. stratified random sampling
 C. Non probability sampling
 D. cluster sampling

14. Statistician prefer an interval estimate instead of point estimate, because of _____
 A. A sample statistic is different from the population mean due to sampling error
 B. There is some doubt on the accuracy of point estimates
 C. A degree of confidence (usually a percent) can be assigned before an interval estimate is made
 D. All of the above

15. A test is performed to decide whether the postulated hypothesis be accepted or not and this involves certain amount of risk. This amount of risk is termed as _____
 A. Rejection area
 B. A level of significance
 C. Critical value
 D. Acceptance Area

16. Suppose a contractor wishes to lower heating bills by using a special type of insulation in houses. If the average of the monthly heating bills is \$78. How one can formulate hypothesis about heating costs with the use of insulation?

$$\frac{10-4}{16} = \frac{6}{16} = \frac{3}{8}$$

$$\frac{4}{36} = \frac{1}{9}$$

$$\frac{4}{20} = \frac{1}{5}$$

R. $\frac{40}{75} = \frac{8}{15}$

$$\frac{40}{75} = \frac{8}{15}$$

$$P(M|NV) = \frac{P(M \cap NV)}{P(NV)}$$

$$P(M|NV) = \frac{40}{75} = \frac{8}{15}$$

$$\frac{40}{55} = \frac{8}{11}$$

$$\frac{40}{75} = \frac{8}{15}$$

$$\frac{40}{75} = \frac{8}{15}$$

$$\frac{40}{75} = \frac{8}{15}$$

$$\frac{40}{75} = \frac{8}{15}$$

$$\frac{40}{75} = \frac{8}{15}$$

$$\frac{40}{75} = \frac{8}{15}$$

D. $H_0: \mu < 78$ and $H_1: \mu \neq 78$

A. $H_0: \mu = 78$ and $H_1: \mu \neq 78$

B. $H_0: \mu = 78$ and $H_1: \mu > 78$

C. $H_0: \mu = 78$ and $H_1: \mu < 78$

17. Suppose that an alternative hypothesis is $H_1: \mu > 50$ and the mean of a sample is $\bar{X} = 52$. If the R software printed a P-value of 0.0356 for a statistical test, then

A. The probability of getting a sample mean of 52 or greater is 0.0356 and H_0 will be rejected at $\alpha = 0.05$

B. The probability of getting a sample mean of 52 or greater is 0.0712 if the true population mean is 50 and H_0 will be rejected at $\alpha = 0.05$

C. The probability of getting a sample mean of 52 or less is 0.0356 and H_0 will be rejected at $\alpha = 0.05$

D. The probability of getting a sample mean of 52 or less is 0.0712 if the true population mean is 50 and H_0 will be rejected at $\alpha = 0.05$

18. What will be approximated sample size with p unknown, margin of error (E) = 0.04 and 95% confidence level?

A. 524

B. 601

19. What type of data do you need for a chi-square test?

A. Qualitative/categorical

B. Quantitative

C. Continuous

D. All of the above

20. Under the Sign Test, if the test value is less than or equal to the critical value

A. Null hypothesis is not rejected

B. Null hypothesis is rejected

C. No decision

D. A & C are answers

21. A nonparametric statistic that uses ranks to determine if there is a relationship between two variables.

A. Runs test

B. Spearman

coefficient

correlation

C. Sign test

D. Kruskal-Wallis test

22. Ten university students were selected and asked how many soft drinks they drink over a two-week period. These students were asked to replace some of the soft drinks with water in order to cut down on the amount of soft drinks that they consumed. At $\alpha = 0.10$, was there a decrease in the amount of soft drinks consumed over a two-week period? The results are shown. (Use critical value at $n=8$ and $\alpha = 0.10$ is equal to 6)

Student	A	B	C	D	E	F	G	H	I	J
Before	6	12	15	20	18	24	9	7	26	21
After	8	10	12	17	14	21	11	8	23	17

A. The test value=10 and we do not reject the hypothesis

B. The test value=7 and we do not reject the hypothesis

C. The test value=-7 and we reject the hypothesis

D. The test value=10 and we reject the hypothesis

23. In hypothesis testing, rejecting null hypothesis while it True is?

- A. Correct decision
 - B. Type I error and its size is α
 - C. Type II error and its size is β
 - D. Type I error and its size is β
 - E. Type II error and its size is α
24. In hypothesis testing, the hypothesis tentatively assumed to be true is
- A. the alternative hypothesis
 - B. either the null or the alternative
 - C. the null hypothesis
 - D. None of the above
25. To make decision in any hypothesis testing problem, calculated value of the test statistic will be compared with _____
- A. Level of significance
 - B. P-value
 - C. Critical/tabulated value
 - D. Degrees of freedom
 - E. None
26. A result is called "statistically significant" whenever" _____
- A. The null hypothesis is failed to reject.
 - B. The p-value is less to the significance level.
 - C. The p-value is less to calculated value of test statistic.
 - D. The p-value is larger than the significance level.
 - E. None
27. Suppose that an investigator would like to compare effectiveness of two possible treatments for hypertension. Thirty-two subjects suffering from hypertension were recruited to the study, with sixteen being randomly allocated to each of treatment. Blood pressure measurements were made on each subject after treatment, assuming that data on effectiveness follows normal distribution and equal variances for two drug type, which one of the following method is appropriate to test whether or not the average blood pressure of patients is equal by drug types, _____
- A. one-sample t-test
 - B. Independent sample t-test.
 - C. Paired sample t test
 - D. One way ANOVA
 - E. Wilcoxon rank sum test
28. The coefficients of variation on the yields of Barely crop from two varieties, variety A and Variety B are 21.3% and 16.57%, respectively, then which of the two variety shows highest variation in yields of Barely crop?
- A. Variety A

- B. Variety B
C. Both variety A and Variety B
D. Can't be decided
E. None
29. If the t ratio for the slope of a simple linear regression equation is -2.48 and the critical values of the t distribution at the 1% and 5% levels, respectively, are 3.499 and 2.365 , then the slope is 2 mgi 5 Significant
- A. Not significantly different from zero.
B. Significantly different from zero at both the 1% and the 5% levels.
C. Significantly different from zero at the 1% level but not at the 5% level.
D. Significantly different from zero at the 5% level but not at the 1% level.
30. The correlation coefficient is used to determine
- A. A specific value of the y -variable given a specific value of the x -variable
B. A specific value of the x -variable given a specific value of the y -variable
C. The strength of the relationship between the x and y variables
D. All of the above
31. In multiple regression analysis, the ratio mean square of residual and mean square error i.e. MSR/MSE yields the
- A. t -test statistic for testing each individual regression coefficient
B. F -test statistic for testing the validity of the regression equation
C. Multiple coefficient of determination
D. Adjusted multiple coefficient of determination
32. In a regression analysis if $SSE = 200$ and $SSR = 300$, then the coefficient of determination is $\frac{200}{500}$
 0.4
- A. 0.6667
B. 0.6000
C. 0.4000
D. All of the above
33. Multiple regression analysis is used when
- A. There is not enough data to carry out simple linear regression analysis.
B. The dependent variable depends on more than one independent variable.
C. One or more of the assumptions of simple linear regression are not correct.
D. The relationship between the dependent variable and the independent variables cannot be described by a linear function.
34. The Durbin-Watson statistic is used to test for
- A. Multicollinearity
B. Heteroskedasticity.
C. Autocorrelation

D. All of the above are correct.

35. When the independent variables are correlated with one another in a multiple regression analysis, this condition is called _____

- A. Heteroscedasticity
- B. Homoscedasticity
- C. Multicollinearity
- D. Normality

36. When testing $H_0: \beta = 0$ against $H_a: \beta \neq 0$ in simple regression, the R output gave the p-value as 0.014. It is appropriate for the researcher to

- A. Conclude that $H_0: \beta = 0$ is not rejected with probability 0.014
- B. Reject $H_0: \beta = 0$ at $\alpha = 0.05$
- C. Conclude that $H_a: \beta \neq 0$ is not rejected with probability 0.014
- D. Reject $H_a: \beta \neq 0$ at $\alpha = 0.05$

37. For the following multiple regression model: $\hat{y} = 2 - 3x_1 + 4x_2 + 5x_3$, a unit increase in x_1 , holding x_2 and x_3 constant, results in _____

- A. An increase of 3 units on average in the value of y .
- B. A decrease of 3 units on average in the value of y .
- C. An increase of 8 units in the value of y .
- D. None of the above.

38. Which of the following linear transformation indicates the transformation of the function $y = \beta_0 e^{\beta_1 x}$

- A. $Y' = \log \beta_0 + \beta_1 x'$, where $Y' = \log y$ and $x' = \log x$
- B. $Y' = \log \beta_0 + \beta_1 x$, where $Y' = \log y$
- C. $Y' = \beta_0 - \beta_1 x'$, where $Y' = \frac{1}{y}$ and $x' = \frac{1}{x}$
- D. None of the above

$$y = \beta_0 e^{\beta_1 x}$$

$$\ln y = \ln \beta_0 + \beta_1 x$$

$$\ln y = \ln \beta_0 + \beta_1 x$$

Use the following result to answer the question 39 and 40

The model is fitted using the software R and the following summary output is obtained.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	???	0.1960	8.438	3.57e-13
x1	5.3036	2.5316	???	0.038834
x2	4.0336	2.4796	1.627	0.107111
x3	-9.3153	2.4657	-3.778	0.000276
x4	0.5884	2.2852	0.257	0.797373

Residual standard error: 1.892 on 95 degree of freedom

Multiple R-squared: 0.1948,

F-statistic: 5.745 on 4 and 95 degree of freedom, p-value: 0.0003483

39. What is the estimate of β_0 ?

$$\beta_0 =$$

$$t = \frac{\beta_0}{S.E.}$$

$$8.438$$

$$0.196$$

$$8.4$$

- A. 0.324
- B. 43.051
- C. 1.654
- D. 1.591

40. How many observations are in the data set?

- A. 99
- B. 100
- C. 95
- D. 96

41. Which statement concerning SPSS Statistics application windows is correct?

- A. Data Editor and session windows are two basic windows in SPSS
- B. Output Viewer and Worksheet windows are two basic windows in SPSS
- C. It is commonly known programming language software
- D. Data Editor and Output Viewer windows are two basic windows in SPSS
- E. None

42. Which menu would you select to run statistical procedures in SPSS?

- A. Graph menu
- B. Data menu
- C. Analyze menu

- D. Transform menu
- E. Stat

43. Which menu item in SPSS is used to create bar graphs and scatter plots?

- A. Graphs menu
- B. Data menu

- C. Transform menu
- D. Analyze menu

44. Which menu item contains the split file and merge files by add cases/variables command?

- A. Graph menu
- B. Data menu

- C. Transform menu
- D. Analyze menu

45. Which of the following Window in SPSS is used for entering and viewing data values?

- A. Data view
- B. Variable view

- C. Output viewer
- D. Syntax editor

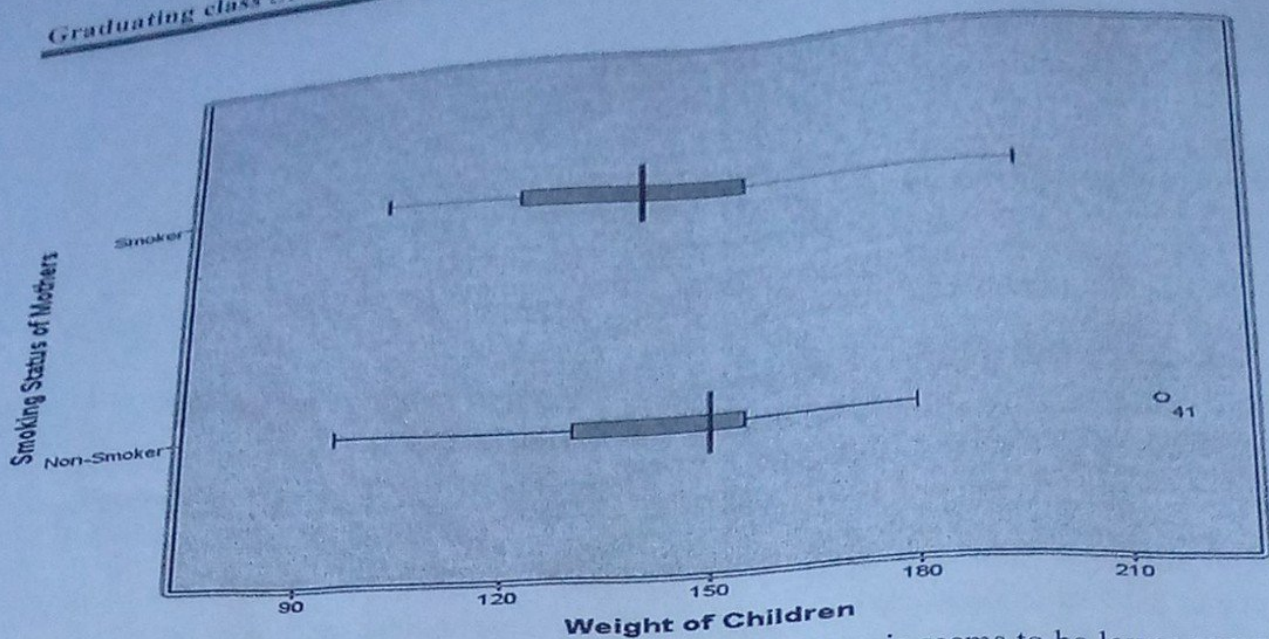
46. Which of the following Window in SPSS is used for creating and defining various characteristics of variables?

- A. Data view
- B. Variable view
- C. Output viewer
- D. Syntax editor

47. To fit simple and multiple linear regression model, which set of instructions, should you give SPSS?

- A. Analyze => Descriptive Statistics => Descriptives.. => select variables => click on OK
- B. Analyze => Descriptive Statistics => Frequencies.. => select variables => click on OK
- C. Analyze => Compare Means => One Way ANOVA => Spearman => click on OK

- D. Analyze=>Correlate=>Bivariate=>select variables=>Pearson=>click on OK
E. Analyze=>Regression=>Linear=>select dependent and independent variables =>click on OK
48. To import an EXCEL file: which set of instructions should you give in SPSS?
A. File=>open =>data then change files of type to excel (.xls)
B. File=>open=> data then change files of type to text (.txt)
C. File=>save as=>choose file directory and type file name =>click ok
D. File=>read text data and change files of type to text (.txt)
E. none
49. How would you use the drop-down menus in SPSS to generate a frequency table?
A. Analyze=>Descriptive Statistics=> Descriptives=> select variables...
B. Analyze=>Descriptive Statistics=>Frequencies=>select variables....
C. Graphs=>Legacy Dialogs=>Bar=>select Simple=> select variables.....
D. Graphs=>Legacy Dialogs=>Scatter/Dot..=>select Simple=> select variables....
E. Analyze=>Descriptive Statistics=>Explore=>select variables.....
50. Which one of the instruction is used to conduct Chi Square test of independence in SPSS?
A. Analyze=>Descriptive Statistics=>Frequencies=> select variables=>Percentages....
B. Analyze=>Descriptive Statistics=>Crosstabs..=> select variables=>Statistics=>chi-square.....
C. Analyze=>Descriptive Statistics=> Crosstabs..=> select variables=>Statistics=>Total Percentages
- D. Analyze=>Descriptive Statistics=>Crosstabs...=> select variables=>Cells=>chi-square....
E. Analyze=>Descriptive Statistics=>Descriptives...=> select variables=>chi-square
51. Which statement is true about the following comparative box plot?



- A. The median weight of children from non-smoker mothers is seems to be lower compared to smoker mothers.
- B. The variability of weights of children from non-smoker mothers is seems to be lower compared to smoker mothers.
- C. There is outlier in the weights of children from both smoker and non-smoker mothers on the upper end of the distributions.
- D. The data on the weights of children from non-smoker mothers seems to normal, whereas from smoker mothers are skewed
- E. None
52. To generate a Pearson's ρ test and coefficients, which set of instructions, should you give SPSS?
- A. Analyze=>Crosstabs=>Descriptive Statistics=> then select variables=>Spearman=>click on OK
- B. Graphs=>Frequencies=> then select variables=>Spearman=>Click on OK
- C. Analyze=>Compare Means=> One Way ANOVA=>Spearman=> click on OK
- D. Analyze=>Correlate=>Bivariate=>select variables=>Pearson=>click on OK
- E. Analyze=>Regression=>Linear=>select variables=>click on OK
53. Different procedures used to compare; type of fertilizer in agronomy, or different temperatures in a reactor vessel in chemical engineering are called _____
- A. Treatment
- B. Measurement unit
- C. Experimental error
- D. Response
54. The random variation present in all experimental results is called _____
- A. Factors
- B. Measurement unit

- C. Experimental error
55. Which principles of design of experiment are used to control extraneous source of variation?
A. Randomization
B. Replication
C. Blocking
D. Control
56. ANOVA is a statistical method of comparing the _____ of several populations.
A. Variances
B. Means
C. Standard deviations
D. All of the above
57. When is completely randomized design applied?
A. When the experimental material is homogenous
B. When the experimental design is heterogeneous
C. A design that is used to control two nuisance sources of variability
D. None of the above
58. The sum of squares treatment is used to measure _____.
A. Variation between groups of samples
B. Variation within group of samples
C. Overall variation in the data
D. None of the above
59. The statistic that compares the variation among the sample means to the variation within the samples is _____.
A. t- test
B. χ^2 -test
C. z- test
D. F- test
60. What would you use to determine whether significant differences were observed between all levels of your independent variable?
A. Box plots
B. Post-hoc tests
C. F-test
D. Descriptive statistics
61. What are two sources of variation among the n observations obtained from completely randomized design trial?
A. Variation due to treatment and variation due to plots
B. Variation due to experimental error and variation due to plots
C. Variation due to treatment and variation due to experimental error
D. None of the above
62. A design that is used to control two nuisance sources of variability is called _____.
A. CRD
B. LSD
C. RCBD
D. BIBD
63. Which of the following is true about Latin square design?

- A. Number of rows are not equal
 B. Number of columns are not equal with number of rows
 C. Number of treatment are equal with both number of columns and rows
 D. Number of treatments are not equal with number of rows

Based on the ANOVA table below answer the questions from 64 to 70 64 - 68

Source	Of	Degree Of	Sum of squares	Mean squares	F-value
Variation		Freedom		?????	
Treatment		??????	864.1	530	?????
Error		??????	6360		
Total		14			

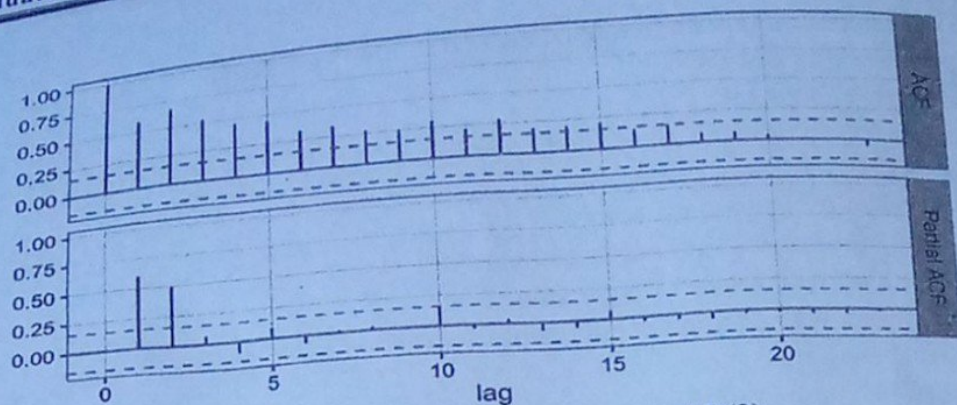
64. The degree of freedom for treatment is _____
 A. 2
 B. 3
 C. 12
 D. 14
65. What is mean square of treatment?
 A. 7224.1
 B. 432.05
 C. 534
 D. 0.81
66. The F-test statistic is _____
 A. 224.1
 B. 0.34
 C. 0.3205
 D. 0.81
67. Factorial designs allow us to study both _____ effects of the independent variables on the dependent variables.
 A. Main and interaction
 B. Dependent and independent
 C. Mean and variance
 D. None of the above
68. Analysis of covariance is a blending of _____ and _____.
 A. Correlation and regression
 B. Regression and ANOVA
 C. Correlation and ANOVA
 D. None of the above
69. What are the core elements of a dissertation/ research?
 A. Introduction; Data Collection; Data Analysis; Conclusions and Recommendations.
 B. Executive Summary; Literature Review; Data Gathered; Conclusions; Bibliography.
 C. Research Plan; Research Data; Analysis; References.
 D. Introduction; Literature Review; Research Methodology; Results; Discussions and Conclusions
70. _____ is applied research that treats knowledge as a form of power and abolishes the line between research and social action.
 A. Action Research
 B. Social Impact Assessment Research
 C. Evaluation Research
 D. Basic Research
71. Which one among the following statements is **false** in the context of descriptive research?
 A. It requires a more developed idea about a social phenomenon.

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- B. It focuses on "how" and "who" questions.
C. It attempts to seek answers to research hypotheses or problems.
D. It summarizes the characteristics of particular individuals or groups.
72. A research design should provide detailed information about _____
A. The study population.
B. The data collection methods.
C. Ethical Issues.
D. All of the above
73. Which one of the following is **correct** about the *Harvard Referencing* style for an article?
A. Author(s)' Surname followed by initials: Title of article. Name of Journal, Volume, number, page numbers of article, year.
B. Author(s)' Surname followed by initials. (Year): Title of article. Name of Journal, Volume, number, page numbers of article.
C. Author(s)' Surname followed by initials. Title of article. Name of Journal. Year; Volume(number): page numbers of article.
D. Author(s)' Surname followed by initials. Title of article. Year. Name of Journal; Volume(number): page numbers of article.
74. How is random sampling helpful?
A. Reasonably accurate
B. An economical method of data collection
C. Free from personal biases
D. All of the above
75. Which one is called non-probability sampling?
A. Quota sampling
B. Cluster sampling
C. Systematic sampling
D. Stratified random sampling
76. A manager associated each employee's name with a number on one ball in a container, then drew balls without looking to select a sample of 5 employees. **What type of sample is this?**
A. Simple random sample
B. Stratified random sample
C. Systematic random sample
D. None of the above
77. Which of the following aspects is a disadvantage of questionnaires?
A. Low response rate
B. Researchers cannot usually observe the respondents reaction to questions
C. Mail/ online questionnaire is not suitable for illiterate community
D. All of the above
78. The studies in which the same subjects are surveyed at different times over extended periods of time is called _____
A. Trend study
B. Panel study
C. Cohort study
D. Probing study
79. Which one of the following is **not** taken as a factor of sample size determination?
A. Level of precision required in the estimates
B. Level of variability of the variables to be estimated

- C. The sample design used
D. None of the above
80. Which of the following is/are the uses of Pilot Survey?
A. It will make a sensible choice from alternative methods of collecting the data.
B. It gives guidance on the adequacy of the sampling frame from which it is proposed to select the sample.
C. It will give the direction on the probable cost and duration of the main survey.
D. All of the above
81. If data for a time series analysis is collected on an annual basis only, which component may be ignored?
A. Trend
B. Seasonality
C. Irregular
D. Cyclic
82. In autoregressive models _____?
A. Current value of dependent variable is influenced by current values of independent variables
B. Current value of dependent variable is influenced by current and past values of independent variables
C. Current value of dependent variable is influenced by both past and current values of dependent variables
D. None of the above
83. A common method known as ratio-to-trend analysis is used to _____
A. Deseasonalize data
B. Take moving average
C. Remove multicollinearity
D. Represent graphical curve
84. What does autocovariance measure?
A. Linear dependence between multiple points on the different series observed at different times
B. Quadratic dependence between two points on the same series observed at different times
C. Linear dependence between two points on different series observed at same time
D. Linear dependence between two points on the same series observed at different times
85. A linear trend was detected for Y, electrical power consumption for the years 1991 ($t=1$) to 2000 ($t=10$). The least squares equation obtained for the trend is $Y = 105.3 + 20.7t$. Forecast the value for 2001.
A. 319.1
B. 212.3
C. 126.0
D. 333.0
86. The time series component that reflects the irregular changes in a time series that are not caused by any other component, and tends to hide the existence of the other more predictable components is called:
A. Trend
B. Cyclical
C. Seasonal
D. random variation

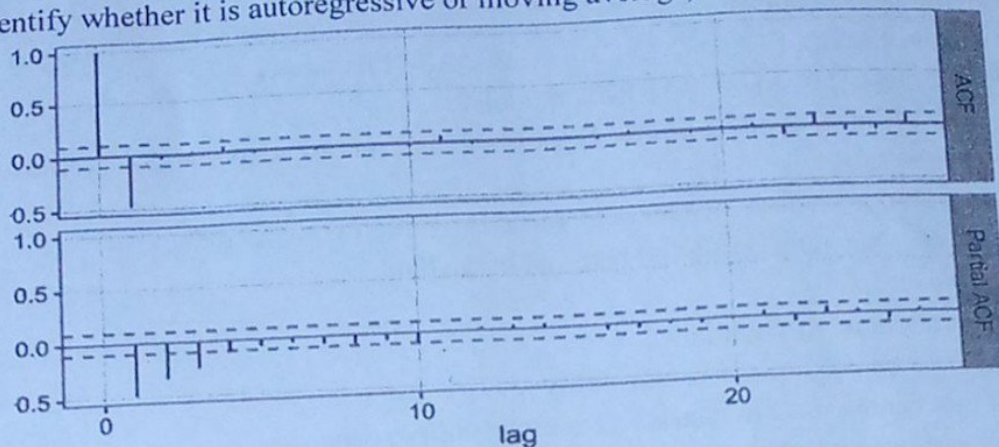
87. Which statement is incorrect about the free-hand method for trend estimation?
- A. It is the simplest and quickest method.
 - B. The method is highly subjective.
 - C. Different persons can fit the same lines for the same data.
 - D. None of the above is
88. Greatest smoothing effect is obtained by using _____
- A. A moving average based on a larger number of periods.
 - B. Exponential smoothing with a small weight value.
 - C. Exponential smoothing with a larger weight value.
 - D. A and B are correct answers.
89. A long run increase or decrease in time series data is known to be _____
- A. Trend variations
 - B. Seasonal variations
 - C. Cyclic variations
 - D. Random/irregular variations
90. Box and Jenkins model building strategy consists _____
- A. Identification (model selection)
 - B. Estimation (model fitting)
 - C. Testing (diagnostic checking)
 - D. All of the above
91. Which one is correct description about ARIMA models?
- A. They are a class of linear models that is capable of representing stationary as well as non-stationary time series.
 - B. ARIMA models do not involve independent variables in their construction.
 - C. They depend heavily on autocorrelation patterns in the data.
 - D. All of the above.
92. What is an AR characteristic polynomial?
- A. A function that can be used to assess normality
 - B. A function of that can be used to characterize invertibility properties.
 - C. A function that can be used to characterize stationary properties.
 - D. All of the above.
93. The augmented Dickey-Fuller unit root test can be used to test for
- A. Normality.
 - B. Independence.
 - C. Stationarity.
 - D. Invertibility.
94. The Yule-Walker equations can be used to _____
- A. Assess if residuals from a stationary ARMA(p, q) fit are normally distributed.
 - B. Calculate residuals for a stationary ARMA(p, q) fit.
 - C. Compute autocorrelations for a stationary ARMA(p, q) process.
 - D. Determine if a variance-stabilizing transformation is necessary.
95. Considering the following ACF and PACF for three time series data. For each series, identify whether it is autoregressive or moving average, and the order (p or q).



- A. AR(1)
B. MA(1)

- C. AR(2)
D. MA(2)

96. Considering the following ACF and PACF for three time series data. For each series, identify whether it is autoregressive or moving average, and the order (p or q).



- A. AR(1)
B. MA(1)

- C. AR(2)
D. MA(2)

97. A researcher studies the factors that determine the number of children future couples decide to have. The regression model to be used will be _____

- A. Poisson
B. Binary logistic
C. Linear regression

- D. ordinal logistic
E. multinomial

98. Which of the following is not a property of a binomial experiment?

- A. The experiment consists of a sequence of an identical trials
B. Each outcome can be referred to as a success or a failure
C. The probabilities of the two outcomes can change from one trial to the next
D. The trials are independent

99. What is the function of a contingency table, in the context of bivariate analysis?

- A. It shows the results you would expect to find by chance

- B. It summarizes the frequencies of two variables so that they can be compared
C. It lists the different levels of the p-value for tests of significance
D. It compares the results you might get from various statistical tests
100. An important application of the chi-square distribution is _____
A. Making inferences about a single population variance
B. Testing for the goodness of fit
C. Testing for the independence of two variables
D. All of the above.
101. Fisher's exact test for a contingency table _____
A. Applies to 2 by 2 tables;
B. Usually gives a larger probability than the ordinary chi-squared test;
C. Is suitable when expected frequencies are small;
D. Is difficult to calculate when the expected frequencies are large.
102. When an odds ratio is calculated from a two by two table _____
A. The odds ratio is a measure of the strength of the relationship between the row and column variables.
B. If the order of the rows and the order of the columns is reversed, the odds ratio will be unchanged.
C. The ratio may take any value.
D. The odds ratio will be changed to its reciprocal if the order of the columns only is changed.
E. The odds ratio is the ratio of the proportions of observations in the first row for the two columns.
103. A significant odds ratio of 2.5 for BMI as a continuous predictor of heart disease in a binary logistic model would indicate which of the following?
A. The odds of heart disease increase by 2.5% for every 1-point increase in BMI.
B. Those with heart disease have 2.5 times higher odds of having an increasing BMI compared to those without heart disease.
C. The odds of heart disease are 2.5 times higher for every 1-point increase in BMI.
D. There are 2.5 times as many people with heart disease as those with higher BMI.
104. Which of the following component is involved in generalized linear models?
A. A systematic component via a linear predictor
B. An exponential family model for the response
C. A link function that connects the means of the response to the linear predictor
D. All of the above
105. Multinomial logistic regression is often considered an attractive analysis because; it does not assume
A. Normality
B. Linearity
C. Homoscedasticity
D. All of the above
- 2.5

- E. None of the above
106. The assumption of proportional odds states that _____
- A. The odds for each explanatory variable must be the same.
 - B. There must be an approximately equal proportion of cases for each level of the ordinal outcome variable.
 - C. The effects of any explanatory variables must be consistent across the different levels of the ordinal outcome variable.
 - D. None of the above
107. Computing the percentage correctly predicted by the model is one way to _____
- A. Assess the statistical significance of parameters.
 - B. Indicates how well a model has predicted group membership.
 - C. Indicates how well the independent variable(s) correlate with the dependent variable.
 - D. Provides a basis for calculating the $\exp(\beta)$ value
108. Ordinal logistic regression is used when _____
- A. The dependent variable is nominal with three or more categories.
 - B. The dependent variable is ordinal with three or more categories.
 - C. At least one independent is nominal with three or more categories.
 - D. At least one independent is ordinal with three or more categories.
109. Which one of the following is not the main objective of demography?
- A. To understand how national and local population estimates are made.
 - B. Introduce and apply the government policy on population study measures.
 - C. Introduce about demographic data and the techniques used in the collection of demographic data.
 - D. Identify demographic variables and their consequences on population change.
 - E. Making prediction and forecasting about the population figure in a give country.
 - F. All of the above
110. What is the consequence of excessive population growth in one country and T.R. Malthus point of view on population growth?
- A. Increase output per worker
 - B. Cause conflict in resource sharing
 - C. Depress the living standards of the masses;
 - D. All except A
111. General fertility rate is a better measure of fertility rate than the crude birth rate because of _____
- A. It eliminates differences in the proportion of females in different population.
 - B. It eliminates the effect of males' composition in the compared population
 - C. It eliminates differences in the proportion of females below 15 ages and male in population.
 - D. It can't indicate differences that really exist.

112. What was the principal driving force of Ethiopian population change in the early years?
- Net migration - excess of out migration over in migration
 - Net migration - excess of in migration over out migration
 - Natural change - excess of births over deaths
 - Natural change - excess of deaths over births
113. What does mean that natural increase in population change in stable population?
- The difference between the number of births and the number of deaths
 - The sum of the number of births and the number of deaths
 - The difference between the number of immigrants and the number of emigrants
 - The sum of the number of immigrants and the number of emigrants
114. Which one of the following demographic tool for examining fertility, mortality and migration in population dynamics?
- Population pyramid
 - Dependency ratio
 - Internal migration measurement
 - Mortality measurement
115. Demographic transition theory states that, as a given country becomes industrialized and civilized, _____
- The death rate declines, but not the birth rate.
 - The birth rate increase and death rate decline.
 - The birth rate declines and then the death rate declines.
 - The death rate declines and then the birth rate declines.
 - Neither birth nor death rates decline.
116. Which one of the following demography study express the future values of a given country population figure by taking into account changes that have occurred in the past?
- Population Dynamic
 - Population Projection
 - Intercensal estimation
 - Postcensal estimation
- Answer questions from 154 to 157 based on the following given information in the table
- Assume that a hypothetical data for a country Y on female population aged between 15 - 49, total birth, female birth and a life table population with radix 100,000 ($5Lx/100,000$) is given in the following table for the year in 1975.

Age	Women	Total Births	Female Births	Age Specific Female Fertility Rate (ASFFR)	$5Lx/100,000$	Female Births to Women in Stationary Population
15-19	977,407	95,157	45,176		0.745	
20-24	916,721	199,144	97,446		0.722	
25-29	977,586	231,213	112,992		0.696	
30-34	888,196	170,901	82,688		0.671	
35-39	755,600	123,546	59,120		0.642	
40-44	620,377	42,863	20,654		0.612	
45-49	438,492	13,501	6,814		0.577	
Total	5,574,379	876,325	424,890			

117. Based on the above given information, what is the gross reproduction of the above population?

- A. 2.1
B. 2.44
C. 2.5
D. 3
E. 3.5

118. Based on the above given information, what is the total fertility rate of the above population?

- A. 5.03
B. 6
C. 2.44
D. 3
E. 3.5

119. Suppose a census that was taken in 1975 enumerated 6490 persons at ages 45 to 49 in a certain area; and 10 years later, in 1985, a second census enumerated 7380 persons at ages 55 to 59 in the same area. During the 10 year period 625 persons died. What is the number of migrants?

- A. 1,515
B. 2,000
C. 890
D. 5,865
E. 265

$$7380 - 6490 = 890 + 625 = 1515$$

120. Consider the following flow matrix of migrants for some selected areas in 1994 census result.

Area of Origin	Area of Destination				
	Sidama	Bale	Gamo-Gofa	Gojjam	Addis Ababa
Sidama		17,273	615	1,191	11,864
Bale	9654		735	1,414	3,500
Gamo-Gofa	995	915		766	17,917
Gojjam	2083	1,163	705		29,698
Addis Ababa	3466	1,622	1,303	2,843	

Based on the given information, what is the out and in migration peoples in Gamo-Gofa, respectively?

- A. 3,358 and 20,593
B. 20,593 and 3,358
C. 995 and 615
D. 915 and 735
E. None of the above